COURSE: BCA-502 (Fifth Semester)

Operation Research

Block-I
Part-I Model Formulation & Graphical Solution

Q1. What is Model Formulation? Explain in Detail.

Introduction

Model formulation is the process of representing a real-life problem in mathematical form so that it can be analyzed and solved using scientific methods. In practical life, organizations face many complex problems such as how much to produce, how to allocate limited resources, how to minimize cost, or how to maximize profit. Since real-world situations involve many variables and constraints, it becomes difficult to solve them directly. Therefore, these problems are simplified and expressed in mathematical equations. This mathematical representation is known as a model. In Operations Research, model formulation is the first and most important step because an incorrectly formulated model will produce incorrect results even if the solution method is correct.

Explanation

The process of model formulation includes the following steps:

1. Identifying Decision Variables
   These are unknown quantities that need to be determined. For example, number of units to produce.
2. Defining Objective Function
   It is the mathematical expression that represents the goal of the problem. The goal may be maximization (profit) or minimization (cost).
3. Identifying Constraints
   Constraints represent limitations such as raw materials, labor hours, machine capacity, or budget.
4. Stating Assumptions
   Assumptions simplify the problem by ignoring minor complexities.

Importance

1. Provides a systematic and logical approach.
2. Helps in optimal decision-making.
3. Saves cost and time.
4. Improves planning and efficiency.

Limitations

1. May oversimplify real-world conditions.
2. Requires accurate data.
3. Difficult for highly complex problems.

Q2. Explain Linear Programming and its Applications in Detail.

Introduction

Linear Programming (LP) is a mathematical technique used to determine the best possible outcome in a given situation subject to certain constraints. The term “linear” means that all relationships between variables are linear (straight-line relationships), and “programming” means planning or optimization. LP helps in making the most efficient use of limited resources. It is widely used in production management, transportation, agriculture, marketing, finance, and military planning.

Explanation

In LP, we try to maximize or minimize an objective function such as profit, revenue, or cost. The objective function and constraints must be linear equations or inequalities.

Example

A factory produces two products X and Y.
Profit per unit:
X = ₹5
Y = ₹3

Objective Function
Maximize $Z = 5x + 3y$

Subject to constraints
$2x + y \leq 100$
$x + y \leq 80$
$x \geq 0, y \geq 0$

Applications

1. Production planning
2. Transportation scheduling
3. Diet problems
4. Advertising budget allocation
5. Inventory management

Advantages

1. Efficient use of resources.
2. Increases profit.
3. Reduces wastage.
4. Scientific decision-making.

Disadvantages

1. Only linear relationships allowed.
2. Assumes certainty.
3. Cannot handle qualitative factors.

Q3. Explain the Structure of a Linear Programming Problem.

Introduction

The structure of a Linear Programming Problem defines how the problem is arranged mathematically. A properly structured LP problem consists of decision variables, objective function, constraints, and non-negativity conditions. Understanding this structure is essential for solving the problem accurately.

Structure

1. Decision Variables
   Represent unknown quantities to be determined.
   Example: $x1$, $x2$.
2. Objective Function
   Represents the goal of the problem.
   Example:
   Maximize $Z = 5x1 + 4x2$
3. Constraints
   Represent limitations.
   Example:
   $2x1 + x2 \leq 100$
   $x1 + 3x2 \leq 90$
4. Non-negativity Condition
   $x1 \geq 0$
   $x2 \geq 0$

Importance

1. Ensures clear representation.
2. Makes solution easier.
3. Avoids ambiguity.

Q4. Explain the Assumptions of Linear Programming in Detail.

Introduction

Linear Programming is based on certain assumptions that simplify real-world complexity. These assumptions make it possible to apply mathematical techniques for solution. If these assumptions are violated, LP may not provide accurate results.

Assumptions

1. Linearity (Proportionality)
   Contribution of each variable to objective function is proportional to its value.
2. Additivity
   Total contribution is sum of individual contributions.
3. Divisibility
   Decision variables can take fractional values.
4. Certainty
   All coefficients are known and constant.
5. Non-negativity
   Decision variables cannot be negative.

Advantages

1. Simplifies mathematical modeling.
2. Makes computation easier.

Limitations

1. Real-life problems may not be linear.
2. Uncertainty ignored.
3. Fractional values may not be practical.

Q5. Explain the General Mathematical Model of Linear Programming Problem.

Introduction

The general mathematical model of Linear Programming represents any LP problem in a standard mathematical form. This form helps in applying solution techniques such as graphical and simplex methods. It provides uniformity and clarity.

General Form

Maximize or Minimize

$Z = c_1x_1 + c_2x_2 + \ldots + c_nx_n$

Subject to

$a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n \leq b_1$
$a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n \leq b_2$
…
$a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n \leq b_m$

Non-negativity

$x_1, x_2, \ldots, x_n \geq 0$

Where

Z = Objective function
x_i = Decision variables
c_i = Profit or cost coefficients
a_{ij} = Resource coefficients
b_i = Resource limits

Advantages

1. Suitable for computer solution.
2. Easy to apply simplex method.
3. Universal representation.

Limitations

1. Only linear constraints allowed.
2. Assumes perfect knowledge of coefficients.

Q6. Explain Constraints and Non-Negativity Conditions in Linear Programming Problem.

Introduction

In a Linear Programming Problem (LPP), constraints represent the limitations or restrictions within which the objective function must be optimized. In real-life situations, resources such as raw materials, labor, machine time, capital, or storage are limited. These limitations are expressed mathematically in the form of linear equations or inequalities called constraints. Along with constraints, non-negativity conditions are imposed to ensure that decision variables do not

take negative values, because negative production, negative labor, or negative quantity has no practical meaning.

Explanation

1. Constraints

Constraints restrict the values that decision variables can take. They are usually written in the form of linear inequalities such as ≤, ≥, or =.

Example

Maximize $Z = 5x + 3y$

Subject to
$2x + y \leq 100$ (raw material limit)
$x + y \leq 80$ (labor limit)

Here, the inequalities are constraints that restrict production.

2. Non-Negativity Conditions

Decision variables must be non-negative.

$x \geq 0$
$y \geq 0$

This ensures feasible and realistic solutions.

Importance

1. Defines feasible region.
2. Makes solution practical.
3. Prevents unrealistic answers.

Q7. Explain Slack and Surplus Variables in Detail.

Introduction

In order to solve an LPP using algebraic methods like the simplex method, all constraints must be converted into equations. Slack and surplus variables are introduced for this purpose. These variables help convert inequalities into equalities without changing the feasible region.

Explanation

1. Slack Variable

A slack variable is added to a “less than or equal to” (≤) constraint.

Example

$2x + y \leq 100$

Add slack variable s1

$2x + y + s1 = 100$

Here, s1 represents unused resource.

2. Surplus Variable

A surplus variable is subtracted from a “greater than or equal to” (≥) constraint.

Example

$3x + y \geq 50$

Subtract surplus variable s2

$3x + y - s2 = 50$

Importance

1. Converts inequalities to equations.
2. Necessary for simplex method.
3. Shows unused or extra resources.

Q8. Explain Standard Form and Canonical Form of LPP.

Introduction

Before solving a Linear Programming Problem using simplex method, it must be converted into a specific format called standard or canonical form. These forms make the problem easier to solve mathematically.

Standard Form

1. Objective function is maximization.
2. All constraints are equations.

3. Right-hand side constants are positive.
4. All variables are non-negative.

Canonical Form

1. Objective function is maximization.
2. All constraints are of $\leq$ type.
3. Non-negativity conditions satisfied.

Importance

1. Ensures uniform representation.
2. Simplifies solution procedure.

Q9. Explain Graphical Solution Method in Detail.

Introduction

The graphical solution method is used to solve Linear Programming Problems that involve only two decision variables. This method provides a visual representation of constraints and feasible region on a graph. The optimal solution is obtained by evaluating the objective function at corner points of the feasible region.

Explanation

Each constraint is converted into an equation and plotted as a straight line on graph. The area that satisfies all constraints simultaneously is called feasible region.

The optimal solution lies at one of the corner points of this region.

Advantages

1. Easy to understand.
2. Visual clarity.

Limitations

1. Only for two variables.
2. Not practical for large problems.

Q10. Explain Feasible Solution, Feasible Region and Optimal Solution.

Introduction

These are fundamental concepts in graphical method.

1. Feasible Solution

Any solution that satisfies all constraints and non-negativity conditions.

2. Feasible Region

The common area that satisfies all constraints simultaneously.

3. Optimal Solution

The feasible solution that gives maximum or minimum value of objective function.

Example

If maximum profit occurs at point (40, 20), then it is optimal solution.

Q11. Explain the Steps of Graphical Method in Detail.

Introduction

The Graphical Method is a technique used to solve Linear Programming Problems (LPP) involving **two decision variables**. It provides a **visual representation** of the feasible region and helps in finding the **optimal solution** (maximum or minimum) of the objective function. The method is particularly useful for understanding the behavior of the LPP and learning the concept of feasible region, corner points, and optimality.

Steps of Graphical Method

1. **Formulate the LPP**
   - Identify **decision variables** (x and y).
   - Define the **objective function** (maximize profit or minimize cost).
   - Identify **constraints** based on resources, capacity, or other limitations.
   - Include **non-negativity constraints** ($x \geq 0$, $y \geq 0$).

   Example:
   Maximize $Z = 5x + 3y$
   Subject to:
   $2x + y \leq 100$
   $x + y \leq 80$
   $x \geq 0, y \geq 0$

2. **Convert inequalities into equations**
    - Replace each inequality with an equality to plot lines.
    - This helps identify the boundary lines of the feasible region.

    Example:
    $2x + y = 100$
    $x + y = 80$

3. **Plot the constraint lines on a graph**
    - Draw the lines on x-y plane.
    - For each line, find two points using intercepts method:
        - Set $x = 0 \rightarrow$ solve for y
        - Set $y = 0 \rightarrow$ solve for x

    Example:
    $2x + y = 100 \rightarrow$ points (0,100) and (50,0)
    $x + y = 80 \rightarrow$ points (0,80) and (80,0)

4. **Determine the feasible region**
    - The feasible region is the **common area** that satisfies all constraints simultaneously.
    - Shade the region that meets all inequalities.
    - Include the non-negativity region ($x \geq 0$, $y \geq 0$).
5. **Identify the corner points (extreme points)**
    - Feasible region forms a polygon.
    - Corner points are **intersections of constraint lines** and axes.
    - These points are crucial because the optimal solution lies at one of them.

    Example:
    For above LPP, corner points: (0,0), (0,80), (20,60), (50,0)

6. **Evaluate the objective function at each corner point**
    - Substitute x and y values of corner points into Z.
    - Calculate the corresponding value of Z for each point.

    Example:
    $Z = 5x + 3y$
    $(0,0) \rightarrow Z = 0$
    $(0,80) \rightarrow Z = 240$
    $(20,60) \rightarrow Z = 220 + 180 = 400$
    $(50,0) \rightarrow Z = 250$

7. **Select the optimal solution**
    - For maximization, choose the point with **highest Z value**.
    - For minimization, choose the point with **lowest Z value**.

Example:
Maximum Z = 400 at point (20,60) → optimal solution.

Conclusion

The graphical method provides a **step-by-step visual approach** to solving LPP. The **optimal solution always lies at a corner point** of the feasible region. Although it is simple and intuitive, it is limited to **two-variable problems** and not suitable for larger problems.

Advantages:

1. Easy to understand and visualize.
2. Provides insight into feasible region and optimal solution.
3. Useful for teaching and small-scale problems.

Limitations:

1. Cannot handle more than two decision variables.
2. Time-consuming for multiple constraints.
3. Not suitable for large industrial problems.

Q12. Why Does the Optimal Solution Occur at Corner Points?

Introduction

In Linear Programming, the feasible region is formed by the intersection of all constraint lines and non-negativity conditions. The **Fundamental Theorem of Linear Programming** states that if an optimal solution exists, it will always occur at a **corner point (extreme point)** of the feasible region. This is because the objective function, being linear, increases or decreases along a straight line, and the maximum or minimum value is reached where it first touches the feasible region.

Explanation

1. Feasible region forms a **convex polygon** in 2D (or polyhedron in 3D).
2. The objective function line ($Z = c1x + c2y$) is linear and moves parallelly.
3. Maximum or minimum value occurs at the first corner it touches while moving in the direction of optimization.

Example

Maximize $Z = 5x + 3y$
Constraints:
$2x + y \leq 100$

$x + y \leq 80$
$x \geq 0, y \geq 0$

Feasible region corner points: (0,0), (0,80), (20,60), (50,0)

Evaluating Z:
(0,0) → Z = 0
(0,80) → Z = 240
(20,60) → Z = 5×20 + 3×60 = 100 + 180 = 280 → **Maximum**
(50,0) → Z = 250

Hence, optimal solution occurs at corner point (20,60).

Conclusion

The linearity of objective function and convexity of feasible region ensures that **optimal solutions lie at extreme points**.

Q13. Explain Multiple Optimal Solutions in Linear Programming

Introduction

Sometimes, a Linear Programming Problem may have **more than one optimal solution**, i.e., multiple points in the feasible region give the same maximum (or minimum) value of the objective function. This occurs when the objective function line is **parallel to a boundary of the feasible region**.

Explanation

1. If objective function is parallel to a constraint boundary:
    - Every point along that boundary gives the same optimal value.
2. Occurs in maximization or minimization problems.

Example

Maximize $Z = 3x + 6y$
Constraints:
$x + y \leq 10$
$2x + y \leq 12$
$x \geq 0, y \geq 0$

Graphically, objective function line is parallel to one constraint: $x + y = 10$
Corner points along this line give same Z.

Conclusion

Multiple optimal solutions provide **flexibility** in decision-making because any point along the boundary yields same result.

Advantages

1. Offers alternative solutions.
2. Provides flexibility in resource allocation.

Q14. Explain Unbounded Solution in Linear Programming

Introduction

An **unbounded solution** occurs in LPP when the feasible region is **open in the direction of optimization**, allowing the objective function to increase (or decrease) indefinitely. In such a case, there is **no finite maximum (or minimum) value**.

Explanation

1. Occurs due to **insufficient constraints**.
2. Feasible region extends infinitely.
3. The objective function keeps improving without limit.

Example

Maximize $Z = x + y$
Constraints:
$x - y \geq 0$
$x \geq 0$

Feasible region extends infinitely along x and y direction → Z can increase without bound → **No finite solution**.

Conclusion

Unbounded solutions indicate that the problem is **improperly constrained** and more realistic restrictions need to be added.

Importance

1. Helps identify missing constraints.
2. Ensures practical feasibility in real-world problems.

Q15. Explain Infeasible Solution in Linear Programming

Introduction

An **infeasible solution** occurs when **no feasible region exists**, i.e., no point satisfies all constraints simultaneously. This usually happens when constraints **contradict each other**.

Explanation

1. Conflicting constraints prevent any common solution.
2. Graphically, feasible region does not exist.

Example

$x + y \leq 5$
$x + y \geq 10$

There is no point (x, y) that satisfies both → **infeasible problem**.

Conclusion

Infeasible solutions indicate a **problem formulation issue**. Constraints must be reviewed for consistency.

Importance

1. Detects unrealistic or contradictory restrictions.
2. Helps refine the model before solving.

Q16. Explain Degeneracy in Linear Programming Problem

Introduction

Degeneracy occurs in a Linear Programming Problem (LPP) when a **basic feasible solution has one or more basic variables equal to zero**. In graphical terms, degeneracy happens when **more than two constraint lines intersect at the same corner point** of the feasible region. It is a special case that is important in the simplex method, as it may cause **repetition of the same solution** in consecutive iterations, potentially leading to cycling.

Explanation

1. In normal cases, **two constraints intersect** to form a corner point.

2. If three or more constraints intersect at the same point, that point is **degenerate**.
3. Degeneracy does **not affect the optimal solution**, but it can increase computational effort.

Example

Constraints:
$x + y \leq 4$
$x \leq 2$
$y \leq 2$

Corner point (2,2) is formed by **three intersecting constraints** → degenerate point.

Importance

1. Helps in understanding special cases in LPP.
2. Important for avoiding cycling in simplex method.

Advantages

1. Solution remains feasible.
2. Provides insight into constraint interactions.

Disadvantages

1. May cause repeated iterations in simplex method.
2. Increases computational complexity.

Conclusion

Degeneracy is a **mathematical special case** where multiple constraints meet at one point. Recognizing it is essential in solving LP problems accurately.

Q17. Explain Redundant Constraints in Linear Programming Problem

Introduction

A **redundant constraint** is a constraint that does not affect the feasible region or optimal solution. It is unnecessary because the feasible region is already bounded by other constraints. Recognizing redundant constraints simplifies problem-solving and reduces computational effort.

Explanation

- If removing a constraint **does not change the feasible region**, it is redundant.

- Redundant constraints may appear due to overly cautious or overlapping restrictions.

Example

Constraints:
$x + y \leq 10$
$x \leq 8$
$y \leq 7$
$x + y \leq 20$

The constraint $\mathbf{x + y \leq 20}$ is redundant because the feasible region is already limited by $x + y \leq 10$, $x \leq 8$, and $y \leq 7$.

Importance

1. Simplifies LPP model.
2. Reduces calculation time in simplex or graphical method.
3. Makes the model clearer and easier to interpret.

Advantages

1. Improves efficiency.
2. Helps avoid confusion.

Disadvantages

1. Ignoring redundancy may complicate calculations.
2. Could make the model unnecessarily large.

Conclusion

Identifying and removing redundant constraints makes solving LPP problems more **efficient and accurate**.

Q18. Explain Bounded and Unbounded Feasible Region

Introduction

In graphical method, the **feasible region** is the area that satisfies all constraints. Based on the constraints, the feasible region can be **bounded** or **unbounded**, which affects whether the LPP has a finite optimal solution.

Bounded Feasible Region

- Completely enclosed by constraints.
- Has finite area and definite maximum/minimum values.

Example:
Constraints forming a triangle or polygon in the first quadrant.

Unbounded Feasible Region

- Open in at least one direction.
- Objective function may increase indefinitely.

Example:
Maximize $Z = x + y$
Subject to $x - y \geq 0$
$x \geq 0$

Feasible region extends infinitely along positive x and y → Z can increase without limit.

Importance

1. Determines existence of finite optimal solution.
2. Indicates adequacy of constraints in problem formulation.

Conclusion

Bounded regions guarantee a **finite optimal solution**, while unbounded regions may lead to **infinite solutions**, requiring careful modeling.

Q19. Explain Limitations of Graphical Method

Introduction

Graphical method is simple and visually intuitive for solving LPP problems. However, it has several limitations that restrict its practical use in real-life scenarios.

Limitations

1. **Limited to Two Variables**
    - Only suitable for 2D problems.
2. **Not Suitable for Large Problems**
    - Problems with more variables or constraints cannot be solved graphically.
3. **Time-Consuming**
    - Manual plotting of multiple constraints is tedious.
4. **Accuracy Issues**

    - Graphs may not provide exact numerical values.
5. **Not Efficient for Computers**
    - Larger industrial problems require algebraic or simplex methods.

Conclusion

Graphical method is **mainly educational**, useful for visual understanding, but limited for large-scale or multi-variable LPP problems.

Q20. Discuss Importance of Linear Programming in Decision Making

Introduction

Linear Programming is a **mathematical optimization tool** used to make **optimal decisions under resource constraints**. It is widely used in business, industry, agriculture, transportation, marketing, and finance. LP helps managers plan effectively, maximize profits, minimize costs, and allocate resources efficiently.

Importance in Decision Making

1. **Optimal Resource Allocation**
    - Ensures best use of materials, labor, and capital.
2. **Profit Maximization**
    - Helps determine the most profitable combination of products or activities.
3. **Cost Minimization**
    - Reduces production, transportation, or operational costs.
4. **Production Planning**
    - Decides quantity of each product to produce.
5. **Inventory Management**
    - Helps in maintaining optimal stock levels.
6. **Marketing Decisions**
    - Allocates budget for advertising, sales, and distribution.

Advantages

1. Promotes **scientific decision-making**.
2. Improves productivity and efficiency.
3. Reduces wastage and unnecessary cost.
4. Provides a **quantitative approach** for complex decisions.

Limitations

1. Only works with **linear relationships**.
2. Assumes **certainty in data**, ignoring uncertainty.

3. Ignores qualitative factors such as human behavior.

Conclusion

Linear Programming is **critical for decision-making** in modern organizations. Despite some limitations, it remains a powerful and widely used optimization tool.

Block-II
Simplex Method and Duality

Q1. What is Simplex Method? Explain its importance in solving Linear Programming Problems.

Answer:
Introduction:
Simplex Method is a **systematic procedure** to solve Linear Programming Problems (LPP) with **more than two variables**, where graphical methods fail. It was developed by **George Dantzig in 1947** and is widely used in business, industry, and operations research. It moves from one feasible solution to another to find the **optimal solution** (maximum or minimum).

Explanation:

1. LPP is first converted into **standard form** using slack, surplus, or artificial variables.
2. The method starts from an **initial basic feasible solution**.
3. **Pivot operations** are performed to improve the objective function iteratively.
4. Iteration continues until **no further improvement** is possible, indicating an optimal solution.

Example:
Maximize $Z = 3x1 + 5x2$
Subject to:
$2x1 + 3x2 \leq 12$
$x1 + x2 \leq 5$
$x1, x2 \geq 0$

Advantages:

- Can handle multiple variables and constraints.
- Guarantees optimal solution if feasible.
- Efficient for computer implementation.

Limitations:

- Requires precise data.
- Computationally heavy for very large LPPs.

Q2. Define Standard Form of Linear Programming Problem with example.

Answer:
Introduction:
Standard form is a **uniform representation** of LPP suitable for applying simplex method. It ensures:

- All constraints are **equations**.
- All variables are **non-negative**.
- Objective function is **maximization**.

Explanation:

- $\leq$ constraints $\rightarrow$ add **slack variables**.
- $\geq$ constraints $\rightarrow$ subtract **surplus variables** and add **artificial variables**.
- $x \geq 0$ for all variables.

Example:
Max $Z = 5x1 + 3x2$
Constraints:
$2x1 + x2 \leq 8 \rightarrow 2x1 + x2 + s1 = 8$
$x1 + x2 \leq 5 \rightarrow x1 + x2 + s2 = 5$
$x1, x2, s1, s2 \geq 0$

Importance:

- Ensures **uniform approach** for simplex method.
- Makes problem compatible with simplex tableau.

Q3. Explain Simplex Algorithm for Maximization Case with an example.

Answer:
Introduction:
Simplex algorithm finds **maximum value** of the objective function systematically by moving from one feasible solution to another.

Steps:

1. Convert LPP to **standard form**.
2. Construct **initial simplex tableau**.
3. Identify **entering variable** (most negative coefficient in Z-row).
4. Identify **leaving variable** (minimum positive ratio of RHS to pivot column).
5. Perform **pivot operation**.
6. Repeat steps 3–5 until **no negative coefficients** in Z-row.

Example:
Max $Z = 3x1 + 5x2$
Constraints:
$2x1 + 3x2 \leq 12$
$x1 + x2 \leq 5$

Use slack variables s1 and s2, construct initial tableau, perform iterations to get **optimal Z**.

Advantages:

- Efficient for multi-variable LPPs.
- Provides exact solution.

Limitations:

- Manual calculation is tedious.
- Requires accurate arithmetic to avoid errors.

Q4. Explain Simplex Algorithm for Minimization Case with an example.

Answer:
Introduction:
Simplex can also solve **minimization LPPs**, usually by converting them into a **dual maximization problem** or using **Big M method**.

Explanation:

1. Convert LPP to **standard form**.
2. Add **artificial variables** for ≥ constraints.
3. Start with initial basic feasible solution.
4. Perform **pivot operations** until minimum is reached.

Example:
Min $Z = 2x1 + 3x2$
Constraints:
$x1 + 2x2 \geq 6$
$x1 + x2 \geq 4$

Advantages:

- Solves minimization problems directly.
- Guarantees minimum cost solution.

Limitations:

- Artificial variables increase computation.
- Slightly more complex than maximization case.

Q5. Explain Multiple Optimal Solutions in Simplex Method with example.

Answer:
Introduction:
Multiple optimal solutions occur when **more than one feasible solution gives the same optimal value**. It happens when the objective function is **parallel to a constraint boundary**.

Explanation:

- If a **non-basic variable has zero reduced cost** at optimal tableau, multiple solutions exist.
- All points along the line segment connecting corner points with same Z are optimal.

Example:
Max $Z = 2x1 + x2$
Constraints:
$x1 + x2 \leq 4$
$x1 \leq 2$
$x2 \leq 3$

Importance:

- Provides flexibility in decision-making.
- Alternative plans can be adopted without changing optimal value.

Q6. Explain Unbounded Solution in Simplex Method with example.

Answer:
Introduction:
Unbounded solution occurs when **feasible region is open in direction of optimization**, allowing objective function to increase indefinitely.

Explanation:

- Occurs due to **insufficient constraints**.
- During simplex iteration, if **all entries in pivot column ≤ 0**, the solution is unbounded.

Example:
Max $Z = x1 + x2$
Constraints:
$x1 - x2 \geq 0$
$x1 \geq 0$

Feasible region extends infinitely → Z can increase without limit → no finite maximum.

DUALITY IN LINEAR PROGRAMMING

Q7. What is Duality in Linear Programming? Explain with importance.

Answer:
Introduction:
Duality is the concept where every **primal problem** has a corresponding **dual problem**. Solving either gives insight into the other.

Explanation:

- Primal maximization → dual minimization.
- Constraints and objective function coefficients are interchanged.

Importance:

- Provides **upper/lower bounds** for optimal value.
- Useful in **sensitivity analysis**.
- Simplifies solution of complex problems.

Example:
Primal: Max $Z = 3x1 + 2x2$
$x1 + x2 \leq 4$
$2x1 + x2 \leq 5$

Dual: Min $W = 4y1 + 5y2$
$y1 + 2y2 \geq 3$
$y1 + y2 \geq 2$

Q8. Formulation of Dual Linear Problem with Example

Answer:
Introduction:
Formulating a dual problem converts **primal constraints** into dual variables and vice versa.

Steps:

1. Convert maximization → minimization or vice versa.
2. Primal constraints → dual variables.
3. Primal variables → dual constraints.
4. RHS of primal → objective coefficients of dual.

Example:
Primal: Max $Z = 3x1 + 2x2$
$x1 + x2 \leq 4$
$2x1 + x2 \leq 5$
$x1, x2 \geq 0$

Dual: Min $W = 4y1 + 5y2$
$y1 + 2y2 \geq 3$
$y1 + y2 \geq 2$
$y1, y2 \geq 0$

Q9. State Standard Results on Duality in LPP.

Answer:

1. **Weak Duality:** For maximization, dual value ≤ primal value.
2. **Strong Duality:** Optimal values of primal and dual are equal.
3. **Complementary Slackness:** If primal constraint not binding, corresponding dual variable = 0.

Importance:

- Confirms correctness of solution.
- Useful in economic interpretation.

Q10. What is Transportation Problem? Explain its importance.

Answer:
Introduction:

Transportation Problem (TP) is a type of LPP that deals with **minimizing cost of transporting goods** from multiple sources to multiple destinations, while satisfying supply and demand.

Example:
Factories F1, F2 supply goods to Warehouses W1, W2. Cost matrix is given. Objective: minimize transportation cost.

Advantages:

- Reduces transportation cost.
- Helps in logistics planning.

Q11. What is Loops in Transportation Table and Their Properties?

Answer:
Introduction:
Loops are **closed paths** in TP used in improving solutions.

Properties:

1. Start and end at same cell.
2. Moves **horizontally or vertically** only.
3. Alternate + and − signs along loop for adjustment.

Importance:

- Used in **MODI or stepping stone method** to reach optimal solution.

Q12. What is Transportation Method?

Answer:
Introduction:
Transportation method is used to **optimize shipping cost** using a **matrix representation** of supply, demand, and cost.

Steps:

1. Construct transportation table.
2. Find initial feasible solution (NWCM, LCM, VAM).
3. Test for optimality using MODI or stepping stone method.

Advantages:

- Systematic approach.
- Handles large problems efficiently.

Q13.What is Linear Programming Formulation of Transportation Problem?

Answer:
Introduction:
TP can be formulated as LP problem:

Variables: x_{ij} = quantity transported from source i to destination j
Objective Function: Min $Z = \Sigma\Sigma C_{ij} x_{ij}$
Constraints:

- Σx_{ij} = supply at source i
- Σx_{ij} = demand at destination j
- $x_{ij} \geq 0$

Q14. What is North-West Corner Method (Initial Solution)?

Answer:
Introduction:
NWCM is a **simple method** to find initial feasible solution starting at **top-left cell**.

Steps:

1. Allocate as much as possible to NW cell.
2. Adjust supply/demand.
3. Move right or down.

Advantages:

- Simple and fast.

Disadvantages:

- May not be optimal.

Q15. What is Least-Cost Method (Initial Solution)?

Answer:
Introduction:
Allocates to the **cell with least cost** first to minimize cost.

Steps:

1. Select minimum cost cell.
2. Allocate max possible units.
3. Adjust supply/demand.
4. Repeat until all supply and demand satisfied.

Advantages:

- Usually closer to optimal solution.

Disadvantages:

- Slightly more computation than NWCM.

Q16. What is Vogel's Approximation Method (VAM)?

Answer:
Introduction:
VAM uses **penalty differences** (between lowest and second lowest cost) to find initial solution **closer to optimal**.

Steps:

1. Compute row and column penalties.
2. Choose row/column with maximum penalty.
3. Allocate to minimum cost cell in selected row/column.
4. Adjust supply/demand and repeat.

Advantages:

- Better initial solution than NWCM and LCM.

Disadvantages:

- More calculation required.

Q17. What is Optimality Test in Transportation Problem?

Answer:
Introduction:
After obtaining an initial solution, check **optimality** using MODI or Stepping Stone method.

Steps:

1. Compute improvement index (Δij) for unused cells.
2. If all $\Delta ij \geq 0 \rightarrow$ solution is optimal.
3. If $\Delta ij < 0 \rightarrow$ adjust allocations using loops.

Q18. What is Degeneracy in Transportation Problem?

Answer:
Introduction:
Degeneracy occurs when **number of allocations < m + n − 1** in TP (m = sources, n = destinations).

Effect:

- Can cause multiple solutions.
- Artificial allocations added to continue method.

Q19. What is Unbalanced Transportation Problem?

Answer:
Introduction:
Occurs when **total supply ≠ total demand**.

Solution:

- Introduce dummy row or column with zero cost to balance.

Importance:

- Allows standard solution methods (NWCM, LCM, VAM).

Q20. What are the advantages and limitations of Transportation Problem?

Advantages:

1. Minimizes transportation cost.
2. Efficient resource allocation.
3. Easy to solve using systematic methods.

Limitations:

1. Assumes linear costs.
2. Ignores qualitative factors.
3. Computational effort increases for large networks.

Block-III
Transportation Problem

Q1. What is a Transportation Problem? Explain its importance.

Answer:
Introduction:
Transportation Problem (TP) is a **special type of Linear Programming Problem (LPP)** used to **minimize the total cost of transporting goods** from multiple sources to multiple destinations while satisfying supply and demand constraints. It is widely used in logistics, supply chain management, and distribution systems.

Explanation:

- Each source has a **limited supply**.
- Each destination has a **fixed demand**.
- **Transportation cost per unit** from each source to each destination is known.
- Objective: find **quantities to transport** from each source to each destination to **minimize total cost**.

Example:

Sources/Destinations	D1	D2	Supply
S1	2	3	50
S2	4	1	30
Demand	40	40	

Objective: Min Z = 2x11 + 3x12 + 4x21 + 1x22

Importance:

- Reduces total transportation cost.
- Helps in efficient resource allocation.
- Assists in planning logistics for industries.

Q2. Explain Loops in Transportation Table and Their Properties.

Answer:
Introduction:
In TP, **loops** are closed paths in the transportation table used to adjust allocations while maintaining feasibility. Loops are crucial during **MODI or stepping stone methods** for checking optimality.

Properties:

1. Loop is **closed**: starts and ends at same cell.
2. Moves **horizontally and vertically only**.
3. Alternate cells in the loop are assigned **+ and − signs** for allocation adjustment.
4. Used for **redistributing units** to reduce total cost.

Example:
In a 3x3 table:
Loop: (1,1) → (1,3) → (3,3) → (3,1) → back to (1,1)

Importance:

- Essential for **finding optimal solution**.
- Maintains supply and demand constraints during reallocation.

Q3. Explain the Transportation Method.

Answer:
Introduction:
Transportation Method is a **matrix-based technique** used to minimize the cost of transporting goods. It converts TP into a **table with costs, supply, and demand**, enabling systematic solution.

Steps:

1. Construct transportation table.
2. Find **initial basic feasible solution (IBFS)** using NWCM, LCM, or VAM.
3. Check **optimality** using MODI method.
4. Adjust allocations along loops to reduce total cost.

Advantages:

- Organised and systematic approach.

- Handles multiple sources and destinations efficiently.

Disadvantages:

- Manual calculations are tedious for large problems.

Q4. What is Linear Programming Formulation of Transportation Probslem?

Answer:
Introduction:
TP can be formulated as an LP problem where **decision variables, constraints, and objective function** are clearly defined.

Formulation:

- **Variables:** x_{ij} = units transported from source i to destination j
- **Objective Function:** Min $Z = \Sigma\Sigma C_{ij} * x_{ij}$ (total transportation cost)
- **Constraints:**
    1. Σx_{ij} = supply at source i
    2. Σx_{ij} = demand at destination j
    3. $x_{ij} \geq 0$ for all i,j

Example:
From previous table, objective: Min $Z = 2x_{11} + 3x_{12} + 4x_{21} + 1x_{22}$
Constraints:
$x_{11} + x_{12} = 50$ (S1 supply)
$x_{21} + x_{22} = 30$ (S2 supply)
$x_{11} + x_{21} = 40$ (D1 demand)
$x_{12} + x_{22} = 40$ (D2 demand)

Importance:

- Standardized formulation allows solving TP using **linear programming methods**.

Q5. Explain North-West Corner Method (NWCM) for Initial Solution

Answer:
Introduction:
NWCM is a **simple method** to find an **initial feasible solution** for TP. Allocation starts from the **top-left cell (north-west corner)** and moves systematically through the table.

Steps:

1. Allocate maximum possible units to NW corner cell without exceeding supply/demand.
2. Adjust remaining supply/demand.
3. Move **right or down** to next cell and repeat until all supply and demand are allocated.

Example:

Sources/Destinations	**D1**	**D2**	**Supply**
S1			50
S2			30
Demand	40	40	

Allocate 40 to (S1,D1), remaining 10 to (S1,D2), etc.

Advantages:

- Simple and fast.
- Provides feasible starting solution.

Disadvantages:

- May not be **cost-optimal**.

Q6. Explain Least-Cost Method (LCM) for Initial Solution

Answer:
Introduction:
LCM selects the **cell with the minimum cost** first for allocation, aiming to reduce total cost from the start.

Steps:

1. Identify cell with least cost.
2. Allocate **maximum feasible units**.
3. Adjust supply/demand and remove fulfilled row/column.
4. Repeat until all supply and demand allocated.

Example:
Cost table:

	D1	**D2**	**Supply**
S1	2	3	50
S2	1	4	30

D1 D2 Supply
Demand 40 40

- Allocate 30 units to (S2,D1) – least cost = 1
- Then 10 units to (S1,D1), etc.

Advantages:

- Usually **closer to optimal solution** than NWCM.

Disadvantages:

- Slightly more calculation required than NWCM.

Q7. Explain Vogel's Approximation Method (VAM) for Initial Solution

Answer:
Introduction:
VAM considers **penalties** (difference between lowest and second lowest cost in each row/column) to prioritize allocation in cells where ignoring cost would increase total expense. It usually produces **near-optimal initial solution**.

Steps:

1. Calculate **row and column penalties**.
2. Choose **row/column with maximum penalty**.
3. Allocate max feasible units to **least cost cell** in selected row/column.
4. Adjust supply/demand and repeat.

Advantages:

- Often provides **initial solution close to optimal**.

Disadvantages:

- More computation than NWCM or LCM.

Q8. Explain Test of Optimality in Transportation Problem (MODI Method)

Answer:
Introduction:

After obtaining an initial feasible solution, **MODI (Modified Distribution Method)** is used to check whether **current solution is optimal**.

Steps:

1. Compute **dual variables U_i and V_j** such that **Cij = Ui + Vj** for occupied cells.
2. Compute **opportunity cost Δij = Cij − (Ui + Vj)** for unoccupied cells.
3. If all Δij ≥ 0 → solution is optimal.
4. If any Δij < 0 → adjust allocation along loop using +/− signs to reduce cost.

Example:
Initial solution obtained using VAM. Calculate Ui, Vj for occupied cells. Compute Δij for empty cells. If Δij negative → move units along loop to reduce Z.

Advantages:

- Systematic way to reach **optimal solution**.
- Ensures minimal total cost.

Disadvantages:

- Requires careful calculation of U, V values.

Q9. Explain Dual of Transportation Model

Answer:
Introduction:
Every TP has a **dual problem** where dual variables (U_i and V_j) represent **shadow prices** of sources and destinations.

Explanation:

- Ui = potential of source i
- Vj = potential of destination j
- Opportunity cost Δij = Cij − (Ui + Vj)
- Helps **interpret economic value** of resources.

Importance:

- Provides **economic interpretation**.
- Identifies valuable sources and costly routes.

Q10. Explain Economic Interpretation of U_i and V_j in MODI Method

Answer:

- Ui = marginal cost or "shadow price" of one unit increase in supply at source i.
- Vj = marginal saving or "shadow price" of fulfilling demand at destination j.
- Δij = Cij − (Ui + Vj) = **additional cost incurred** if 1 unit shipped via unused cell (i,j).

Importance:

- Guides management decisions about **increasing supply or demand**.
- Shows which routes are **economically attractive**.

Q11. Steps of MODI Method

Answer:

1. Find **initial basic feasible solution** (NWCM, LCM, or VAM).
2. Assign **U_i and V_j values** (set one Ui = 0, solve equations Cij = Ui + Vj for occupied cells).
3. Compute **Δij = Cij − (Ui + Vj)** for all unoccupied cells.
4. If all Δij ≥ 0 → solution is optimal.
5. If any Δij < 0 → select most negative Δij, form **loop**, adjust allocations (+/−) along loop.
6. Repeat steps 2–5 until **all Δij ≥ 0**.

Advantages:

- Provides **optimal solution** systematically.

Disadvantages:

- Calculation can be tedious for large tables.